

Lecture 1

1/21/26
JRA

Welcome to Biomedical Imaging!

Today's lecture will present an overview of course mechanics & course's somewhat unconventional subject matter

Course Mechanics - see Syllabus

- Subject matter
 - Microscopy
 - Medical Imaging
 - Image Processing & Analysis
- Office hours
- Grading

Homework	30%	} OK to work together
In-Class activities	10%	
Midterm	25%	- Microscopy
Final	35%	< Medical Imaging Image Processing & Analysis

Subject Matter

Course is devoted to imaging

This is a hot topic driven by advances in computers & other technologies.

(1) Light Microscopy

Light microscopy uses optical radiation to image specimens, living & dead.

Electron microscopy uses high energy electrons to image specimens

- Higher resolution than light microscopy BUT
- Requires a vacuum $\Rightarrow$ dead specimens only

Course covers whole spectrum of light microscopy techniques but does NOT cover electron microscopy

(2) Medical Imaging

- Old days - did exploratory surgery
- Now - image

Imaging is a black box for most people

Knowing how image is formed is empowering, an avenue to great jobs, & helps to avoid mistakes. For example, in 3D microscopy, slice separation in the axial (z) direction should be $\leq \frac{1}{2}$ resolution to avoid sampling errors (like "aliasing") (e.g., 600 nm resolution $\Rightarrow$ sample at 300 nm)

(3) Digital Image Processing & Analysis

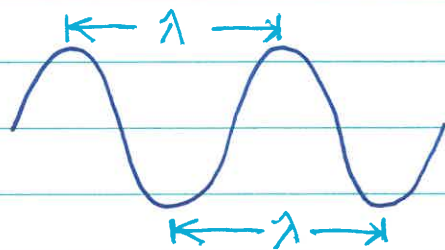
"DIP" is used to clean up or rehabilitate images

Importance of Waves

Everything in class is based on interactions of waves w/ specimen or patient

Electromagnetic (EM) radiation is used w/ most techniques.

EM radiation is characterized by its wavelength, λ



λ = peak-to-peak
(or trough-to-trough)
spacing

Differences in λ give rise to markedly different properties

see initial PowerPoint slide

Decreasing Wavelength / Increasing Energy

Radio Waves - body mostly transparent to radio waves, so signal emerges to give information. Used in MRI.

Infrared (IR) - penetrate ~ 1 cm

Visible - penetrate a few mm

- Bad for imaging body
- Good for imaging cells

Ultraviolet (UV) - penetrates even less

- Kills cells
- Causes cataracts & skin cancer

X-Rays & Gamma Rays - penetrates body

- Some absorbed
- Some emerges to form image
- "ionizing radiation"

ultrasound (US), in contrast, is a sound wave

Common Themes

There are several common themes that will run through the class

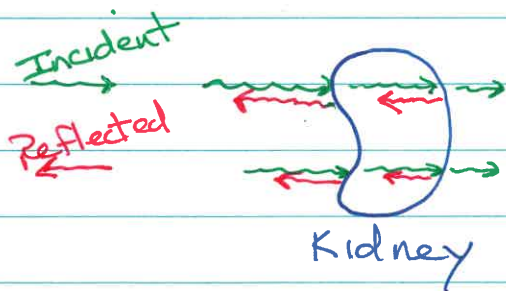
(1) Waves - waves interact w/ matter to form an image

(2) Basis (or mechanism) of image formation

We'll look at ^① how the particular wave (e.g., optical radiation, x-rays, ultrasound) interact w/ specimen/patient, & ^② how the interactions are used to create an image

Example - Ultrasound

Ultrasound uses reflected sound waves to form an image



- Send in incident sound waves
- Detect reflected sound waves (i.e., detect "echoes")

- Echo Time - monitor echo time & use to determine object depth
 - + Surface structure - time for echo to return is short
 - + Deep structures - time for echo to return is long

- Echo Strength - monitor echo strength to determine object brightness

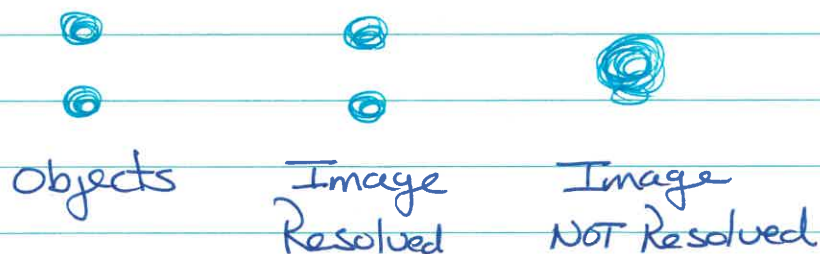
+ Strong echo - bright
+ Weak echo - dim

(3) Image Quality

Look at several measures to characterize image quality, especially resolution & contrast

(3a) Spatial Resolution

Spatial resolution refers to the ability to distinguish objects as distinct, versus having them overlap in image & look like one object



How close can objects get before they cannot be distinguished?
How close & still create distinct images?

- Standard light microscopy

Resolution limit is $\sim 200 \text{ nm}$ ($0.2 \mu\text{m}$)

Severe limitation - objects closer than this cannot be distinguished

Cells are $\sim 10 \mu\text{m}$ in size, so resolution is $\sim 2\%$ of cell size

In the late 1990s, biophysicists developed superresolution light microscopy
Resolution $\approx 20 \text{ nm}$

- Medical imaging

Resolution is a much bigger number than w/ microscopy, but the body is much larger than a cell, so relative resolution is more comparable

Resolution limit is $\sim 0.1 \text{ mm}$ to $\sim 1 \text{ cm}$

$\uparrow$ x-ray

$\uparrow$ nuclear imaging

These techniques all have strengths & weaknesses

(3b) Contrast

Contrast provides a way to distinguish signal variations at different points in an image.

- Grayscale variations - differences in white, gray, black
- Color variations - differences in color (i.e. intensity)

Otherwise, w/o contrast, can't see anything

See PowerPoint slide w/ nerve cell images generated w/ different transmitted light microscopy techniques (all grayscale)

- Brightfield - cells almost invisible
- Darkfield - cells bright on dark background
- Phase contrast - cells dark on bright background

Fluorescence microscopy uses color for contrast

MRI is a powerful technique not because of its resolution, which is standard, but because of its exquisite soft tissue contrast

(3c) Temporal Resolution

How quickly can you collect images?

- Ultrasound - excellent (real-time) temporal resolution (~30 Frames/sec)
(can see beating heart)
- MRI - poor temporal resolution
(very slow)

(3d) Noise

Noise is random signal superimposed on the signal of interest

Nuclear images (generated by collecting radioactivity) are very prone to being noisy

(3e) Structural vs Functional

Does technique yield structural information or functional information?

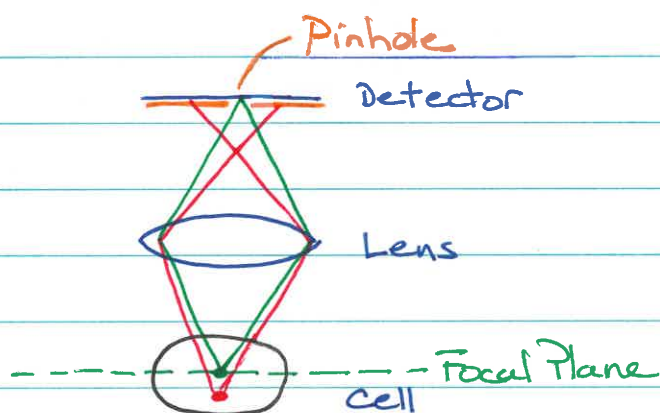
- Structural - is rib broken?
- Functional - is thyroid hypo or hyper active?

Role of Physics

First-year physics allows an understanding of a lot

Example 1 - how does confocal microscopy deblur images?

Confocal microscopy was the brainchild of Marvin Minsky (of artificial intelligence fame) in the 1950s



Green light, from focal plane, in focus at detector & passes thru pinhole

Red light, from below focal plane, in focus before detector & blocked because it misses pinhole

This is why standard fluorescence images are blurry: they are formed from fluorescence originating above & below the focal plane, as well as in the focal plane, contaminating the image

Confocal microscopy  In-focus light makes it thru pinhole & is captured by detector
Out-of-focus light is mostly blocked at pinhole (it's "rejected") & so is NOT captured by detector

Example 2 - how are ultrasound images formed?

Looked at ultrasound simulation showing that imaging requires reflection AND transmission at interfaces (the latter to allow imaging of deeper structures)

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